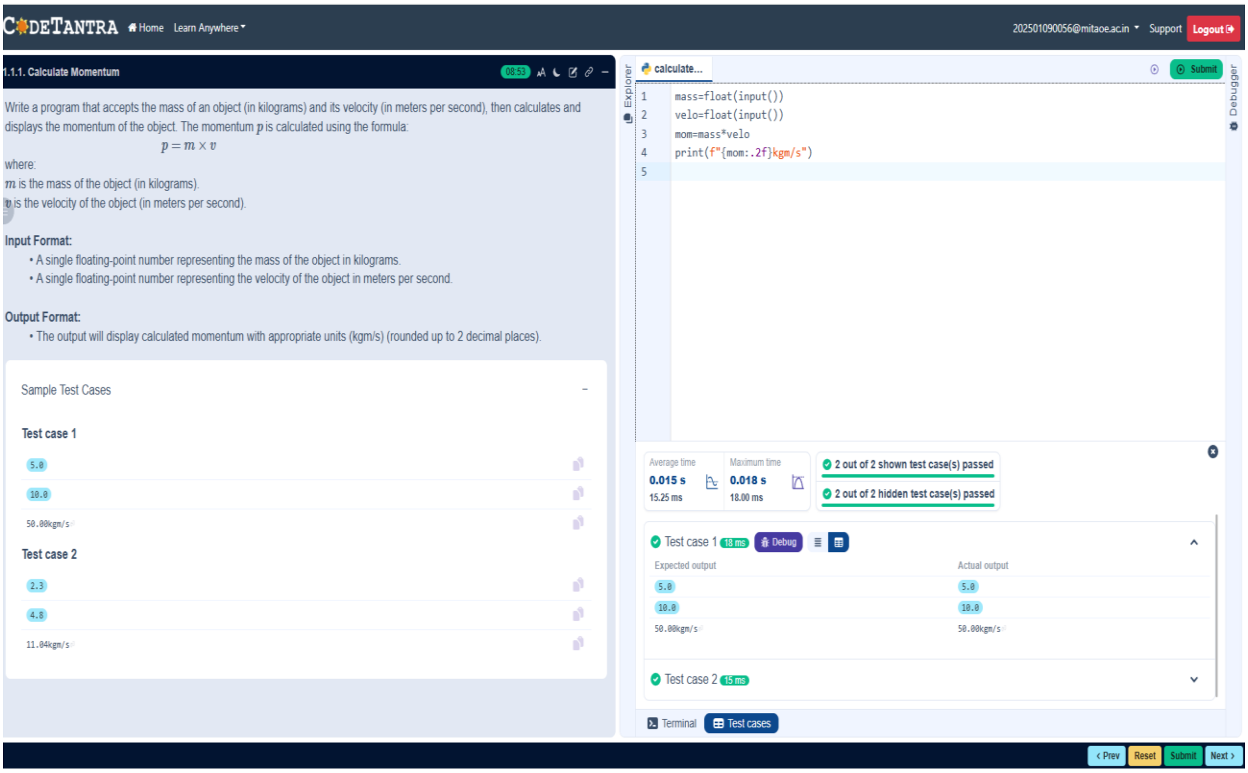
Codetantra Practical 1

Practice Lab Assignment



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.1.2. Conditional Calculation Based on the Number of Digits

Write a Python program that accepts an integer n as input. Depending on the number of digits in n.

Constraints:

$1 \le n \le 999$

Input Format:

- The input consists of a single integer n.

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

Test case 1

9

81

Test case 2

166

5.50

Test case 3

24

4.90

Test case 4

condition...

```
import math
n=int(input())
if 1<=n<=9 :
    print(n**2)
elif 10<=n<=99:
    print(f"{math.sqrt(n):.2f}")
elif 99<=n<=999:
    result=n**(1/3)
    print(f"{result:.2f}")
else :
    print("Invalid")
```

Average time 0.011 s 11.14 ms

Maximum time 0.013 s 13.00 ms

4 out of 4 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 13 ms Debug

Expected output	Actual output
9	9
81	81

Test case 2 12 ms

Test case 3 10 ms

Terminal Test cases

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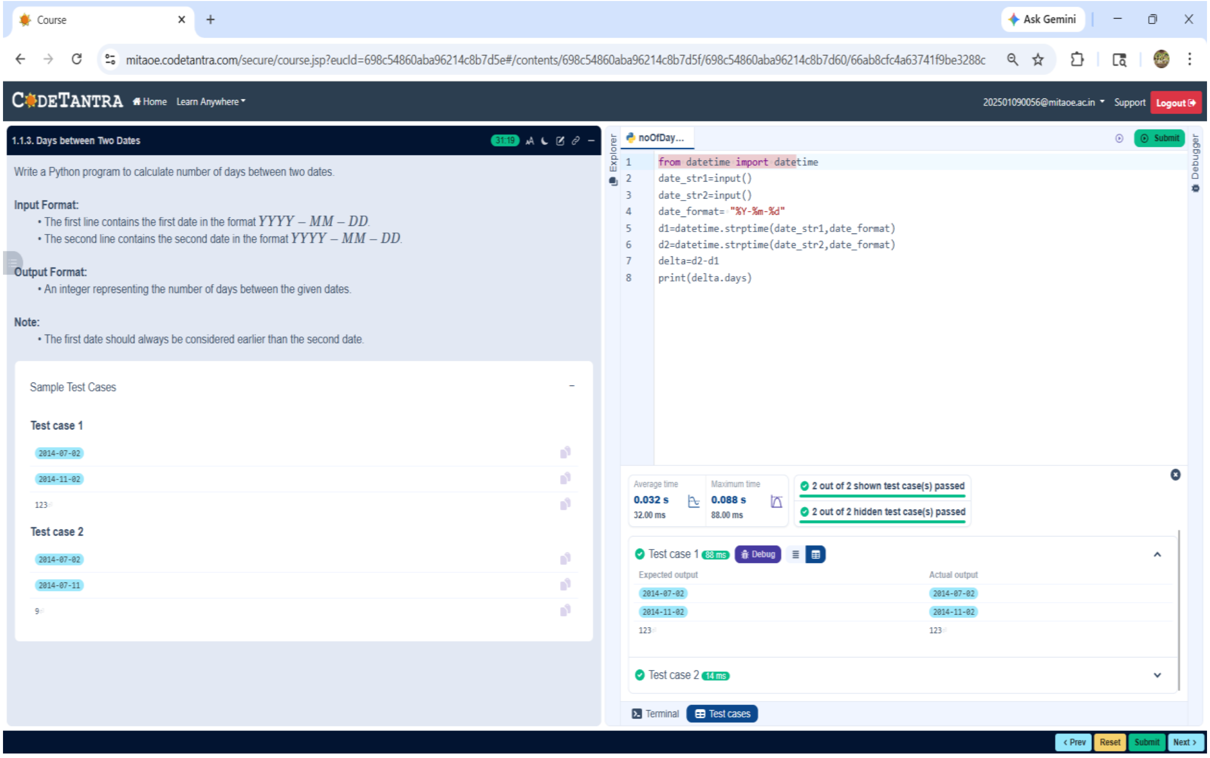
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1.1.3. Days between Two Dates
Write a Python program to calculate number of days between two dates.
Input Format:
• The first line contains the first date in the format YYYY – MM – DD.
• The second line contains the second date in the format YYYY – MM – DD.
Output Format:
• An integer representing the number of days between the given dates.
Note:
• The first date should always be considered earlier than the second date.
Sample Test Cases
Test case 1
2014-07-02
2014-11-02
123
Test case 2
2014-07-02
2014-07-11
9
Explorer
noOfDay...
Submit
Debugger
from datetime import datetime
date_str1=input()
date_str2=input()
date_format= "%Y-%m-%d"
d1=datetime.strptime(date_str1,date_format)
d2=datetime.strptime(date_str2,date_format)
delta=d2-d1
print(delta.days)
Average time
0.032 s
32.00 ms
Maximum time
0.088 s
88.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1 88 ms
Debug
Expected output
2014-07-02
2014-11-02
123
Actual output
2014-07-02
2014-11-02
123
Test case 2 14 ms
Terminal
Test cases
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1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

Test case 1

5367

7635

Test case 2

0132

231

reverseN...

```python

n=int(input())
reversed_num=int(str(n)[::-1])
print(reversed_num)
```

Average time 0.012 s 12.40 ms

Maximum time 0.015 s 15.00 ms

2 out of 2 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 15 ms Debug

Expected output	Actual output
5367	5367
7635	7635

Test case 2 12 ms

Terminal Test cases

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1.1.5. Multiplication Table

03:43

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

```
<number> x <multiplier> = <result>
```

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

Test case 1

```
8
8 x 1 = 8
8 x 2 = 16
8 x 3 = 24
8 x 4 = 32
8 x 5 = 40
8 x 6 = 48
8 x 7 = 56
8 x 8 = 64
```

multiplica... Submit

```python
#write your code here...
n=int(input())
for i in range(1,11):
    print(f"{n} x {i} = {n*i}")
```

Average time	Maximum time
0.008 s	0.010 s
8.25 ms	10.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 10 ms Debug

Expected output	Actual output
8	8
8 x 1 = 8	8 x 1 = 8
8 x 2 = 16	8 x 2 = 16
8 x 3 = 24	8 x 3 = 24
8 x 4 = 32	8 x 4 = 32
8 x 5 = 40	8 x 5 = 40

Terminal Test cases

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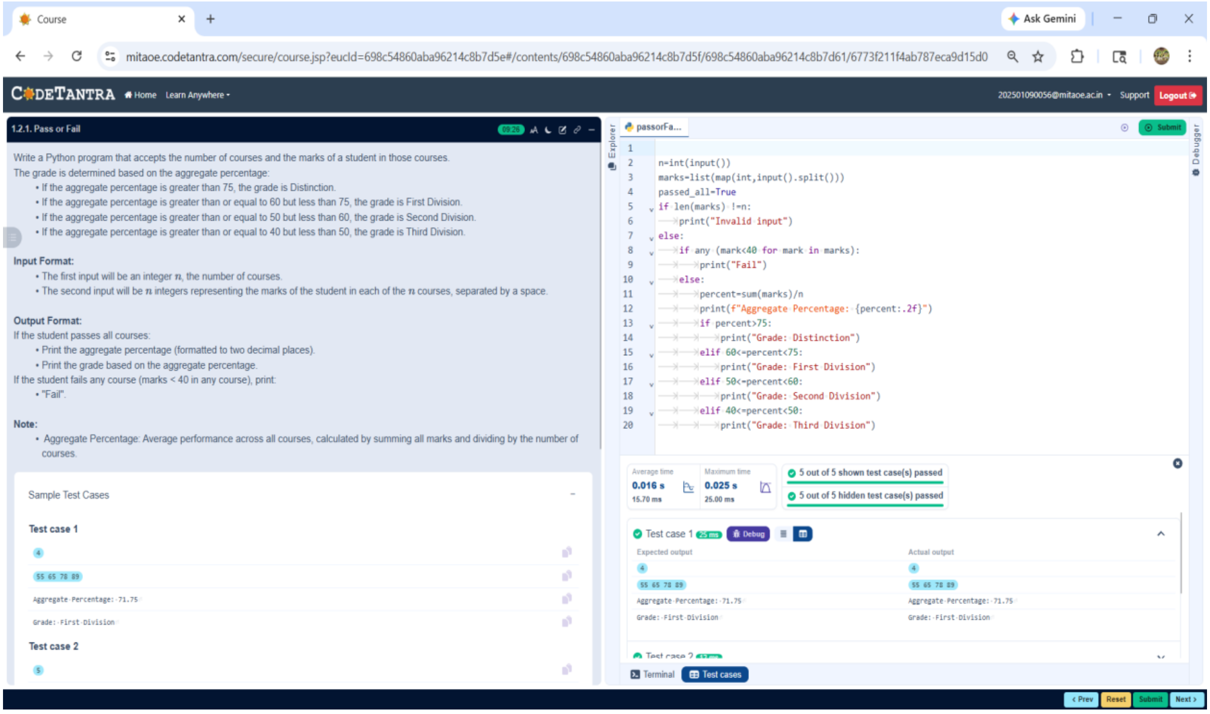
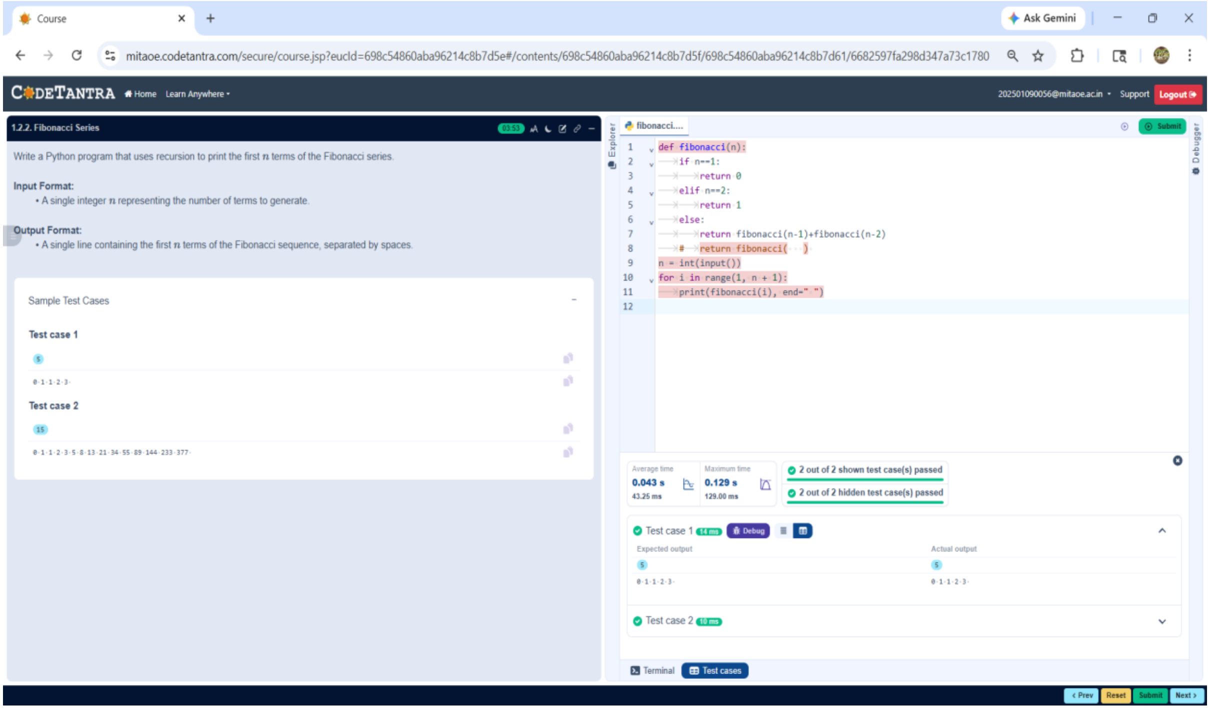
Lab Assignment





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1.2.2. Fibonacci Series
03:53
Write a Python program that uses recursion to print the first n terms of the Fibonacci series.
Input Format:
• A single integer n representing the number of terms to generate.
Output Format:
• A single line containing the first n terms of the Fibonacci sequence, separated by spaces.
Sample Test Cases
Test case 1
5
0 1 1 2 3
Test case 2
15
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377
fibonacci....
Submit
def fibonacci(n):
if n==1:
return 0
elif n==2:
return 1
else:
return fibonacci(n-1)+fibonacci(n-2)
# return fibonacci( )
n = int(input())
for i in range(1, n + 1):
print(fibonacci(i), end=" ")
Average time
0.043 s
43.25 ms
Maximum time
0.129 s
129.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
14 ms
Debug
Expected output
5
0 1 1 2 3
Actual output
5
0 1 1 2 3
Test case 2
10 ms
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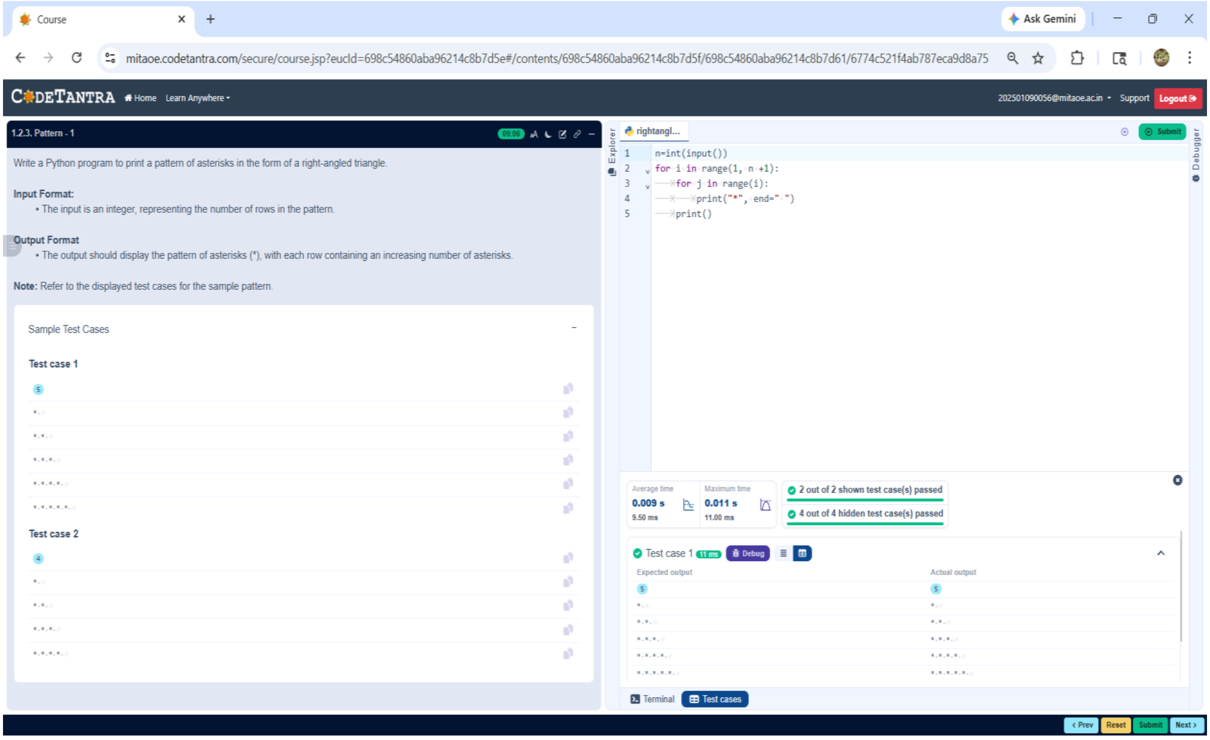
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1.2.3. Pattern - 1
09:06
Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.
Input Format:
• The input is an integer, representing the number of rows in the pattern.
Output Format
• The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.
Note: Refer to the displayed test cases for the sample pattern.
Sample Test Cases
Test case 1
5
*
* *
* * *
* * * *
* * * * *
Test case 2
4
*
* *
* * *
* * * *
rightangl...
Submit
Explorer
Debugger
n=int(input())
for i in range(1, n +1):
for j in range(i):
print("*", end=" ")
print()
Average time
0.009 s
9.50 ms
Maximum time
0.011 s
11.00 ms
2 out of 2 shown test case(s) passed
4 out of 4 hidden test case(s) passed
Test case 1
11 ms
Debug
Expected output
Actual output
5
5
*
*
* *
* *
* * *
* * *
* * * *
* * * *
* * * * *
* * * * *
Terminal
Test cases
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1.2.4. Pattern - 2

04:34

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases

Test case 1

```
5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

Test case 2

```
8
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

numberP...

Submit

```python
n=int(input())
for i in range(1,n+1):
	for j in range(i):
		print(j+1,end=" ")
	print()
```

Average time	Maximum time
0.021 s	0.035 s
21.00 ms	35.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 35 ms Debug

Expected output	Actual output
5	5
1	1
1 2	1 2
1 2 3	1 2 3
1 2 3 4	1 2 3 4
1 2 3 4 5	1 2 3 4 5

Terminal Test cases

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